TrES-3b

R-0345

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_250601` · created 2026-07-18T08:22:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460827.8712 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.81 +/- 2.24 [%]
Orbital Inclination (inc)	83.01 +/- 2.85
Airmass coefficient 1 (a1)	0.086 +/- 0.014
Airmass coefficient 2 (a2)	2.38 +/- 1.46
Scatter in the residuals of the lightcurve fit is	15.82 %
Best Comparison Star	#1 - [441, 408]
Optimal Aperture	1.86
Optimal Annulus	6.41
Transit Duration (day)	0.061 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.061 d vs 0.0567 d (deviation 0.19σ)
Mid-time O–C	3.2 min
Depth SNR	3.4
Residual scatter	15.82 %

Light curve

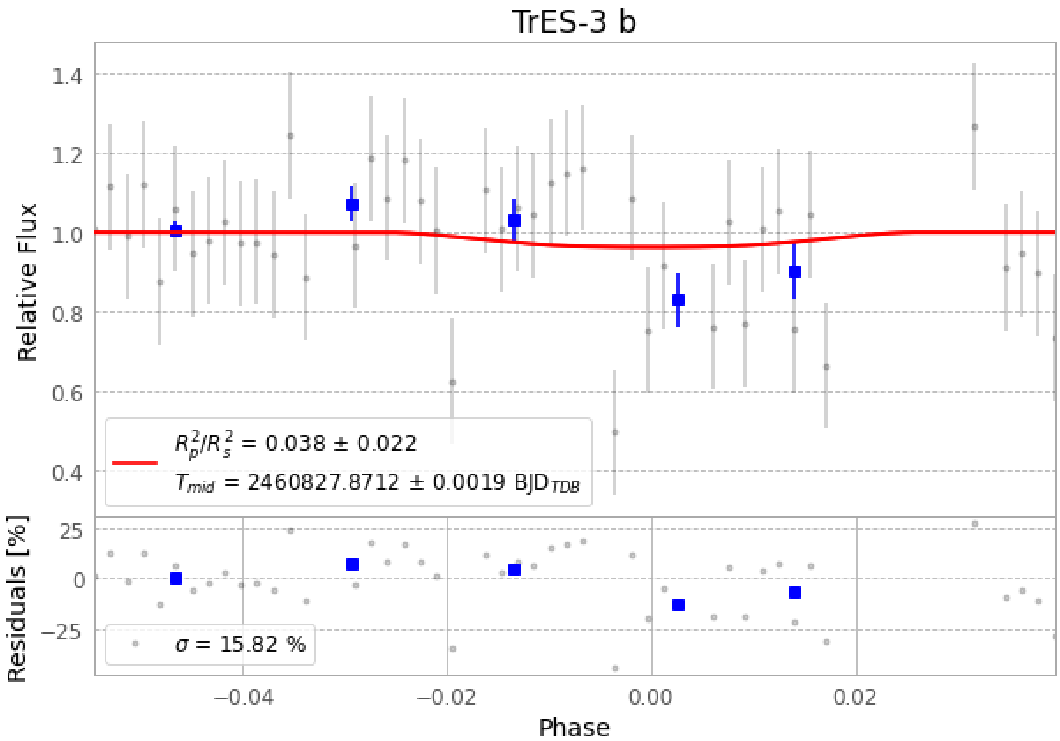

Figure 1 — FinalLightCurve_TrES-3 b_2025-06-01.png (SHA-256 3e931d90a7b85806…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 250], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2c50fe2cff158137…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da`

Data provenance

69 FITS frames (45 MB), `TRES-3250601070618.FITS` → `TRES-3250601103015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250601.log (176bd4479f6f…), FinalLightCurve_TrES-3 b_2025-06-01.png (3e931d90a7b8…), FinalLightCurve_TrES-3 b_2025-06-01.csv (ce9ab85c8c09…)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-18 08:22Z.